App Dev Project Report

1. Student Details

- Name: Atharva Bacchuwar
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2. Project Details

- Project Title: Placement Portal Application (PPA) - V2
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3. AI/LLM Declaration

I used Google DeepMind's Antigravity (Gemini) to assist in refining database schema relationships, designing Vue.js single-page component routing, structuring Celery asynchronous tasks, and refactoring API routes to follow standard RESTful conventions.

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4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend REST API framework

Technology / Library	Purpose
Vue.js (v3 via CDN)	**Reactive frontend Single Page Application (SPA) structure**
Bootstrap 5	**Responsive styling, table design, and dashboard layouts (No other CSS framework used)**
SQLite	**Local database engine for storing users, profiles, drives, and applications**
SQLAlchemy	**Object Relational Mapper (ORM) for python-to-SQL translation and relationship cascade handling**
Redis	**In-memory database used as a broker for Celery and for API response caching**
Celery	**Distributed task queue for asynchronous file generation and scheduled cron/beat jobs**
Werkzeug	**Secure password hashing (`pbkdf2:sha256`) and verification**

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5. Database Schema / ER Diagram

Tables:

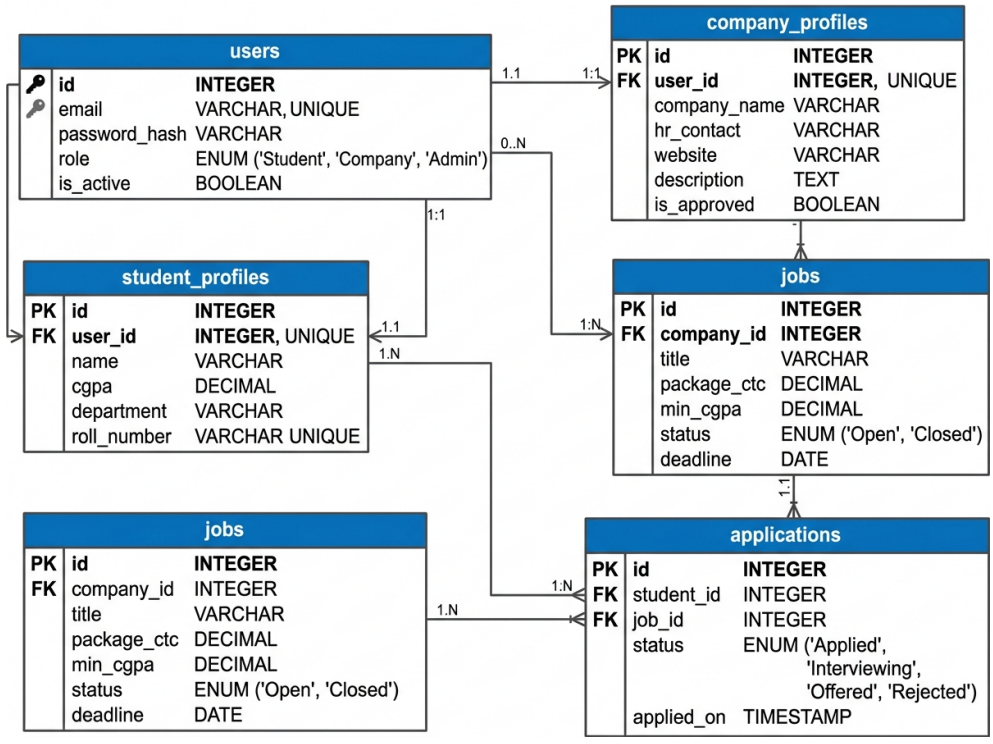
1. User: Holds core credential and authorization state.

- `id` (

Relationships:

- One-to-One: `User.id` <-> `StudentProfile.user_id`
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ER Diagram:



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6. API Resource Endpoints

6.1. Authentication (`/api/auth`)

- `POST /register/student` - Self-register a new Student account.

- `POS

6.2. Student Endpoints (`/api/students`)

- `GET /profile` - Retrieve logged-in student academic details.

- `PUT

6.3. Recruiter Endpoints (`/api/companies`)

- `GET /dashboard` - Fetch recruiter profile and posted drives.

- `POS

6.4. Job & Application Listings (`/api/jobs`)

- `GET /` - Fetch all Admin-approved placement drives. Checks student CGPA against minimum drive requirements and flags eligibility.

- `POS

6.5. Admin Endpoints (`/api/admin`)

- `GET /stats` - Query total stats (percentage of placed students, registration counts). Caching decorator `@cached_api` applied.

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7. Architecture and Features

File Structure:

- `app.py` - Flask API initialization, blueprint registrations, default admin seeding (`admin@iitm.in` / `admin123`), and Celery scheduling definition.

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